MIL-W-16193B
23 September 1977
SUPERSEDING
MIL-W-16193A
22 September 1965

MILITARY SPECIFICATION

WIRE, NONELECTRICAL, PLATINUM ALLOY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers platinum alloy wire for use in direct casting to invested parts of dental prosthetic appliances.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-up.

STANDARDS

FEDERAL

FED-STD-151 - Metal; Test Methods.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th St., Phila., PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6520

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Platinum alloy wire covered by this specification shall be of the following composition:

Gold	-	20.0 - 20.5 percent.
Platinum	-	50.0 - 50.5 percent.
Palladium	-	26.0 - 26.5 percent.
Silver	-	Remainder.

3.2 Description. The platinum alloy wire shall be suitable for use in fabricating wrought prosthodontic clasps, pins, and posts to which precious metal dental alloys can be cast.

3.2.1 Dimensions. Two wires shall be furnished for each unit of issue; each wire shall measure 9 inches, minimum in length, and shall be 0.0400 inch in diameter, plus or minus 0.0005 inch.

3.2.2 Weight. Each unit of issue shall weigh 6.220 grams (4 pennyweight). Individual units of issue of platinum alloy wire shall have a weight tolerance not greater than minus 1.0 percent. Total weight of wire for each delivery to the Government shall have no minus weight tolerance.

3.3 Physical properties.

3.3.1 Ultimate tensile strength. Ultimate tensile strength shall be not less than 120,000 pounds per square inch when given the heat treatment as specified in 4.3.1.

3.3.2 Elongation. The elongation shall be not less than 10 percent as received from the manufacturer, and not less than 10 percent when given the heat treatment as specified in 4.3.1.

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3.4 Finish. The wire shall have a bright, smooth finish.

3.5 Instructions. Adequate instructions for heat treating and using the platinum alloy wire shall be furnished with each package.

3.6 Identification marking. Each length of wire shall be die stamped with an identification or trademark at one end. Ends may be flattened for this purpose.

3.7 Workmanship. Workmanship shall be first class throughout. The platinum alloy wire shall be free from defects which detract from its appearance or impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties). Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.2.1 Examination. Wire shall be examined for defects including, but not limited to, those listed in table I.

TABLE I. Classification of defects.

Categories	Defects
Major:	
101	Diameter of wire not as specified.
102	Workmanship defects present.
103	Finish not as specified.
104	Cracks, laminations, pitted areas present.

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TABLE I. Classification of defects.(continued)

Categories	Defects
Minor:	
201	Units of issue of wire not furnished in 2 lengths.
202	Identification marking not as specified.
203	Instructions missing, incomplete, illegible, or incorrect.
204	Length of wire not as specified.

The inspection levels and acceptable quality levels (AQL's) shall be as listed in table II. The unit of product for examination shall be one unit of issue (4 pennyweight) of wire.

TABLE II. Examination criteria.

Type defect	Inspection level	AQL (Percent defective)
Major	S-2	1.5
Minor	S-2	4.0

4.2.2 Testing. The wire shall be tested as specified in table III and 4.3. The unit of product for testing shall be one unit of issue (4 pennyweight) of wire. Where feasible, the same sample unit shall be used for the determination of 2 or more test characteristics.

TABLE III. Test criteria.

Characteristic	Requirement	Test procedure	Number of units tested per lot
Composition analysis	3.1	4.3.4	1/
Weight, individual	3.2.2	4.3.5	5 2/
Weight, total	3.2.2	4.3.5	Each delivery to Government
Ultimate tensile strength	3.3.1	4.3.2	5 3/
Elongation	3.3.2	4.3.3	5 untreated 3/ 5 hardened 5 softened

1/ A sufficient quantity of alloy used in the formulation of each lot of the platinum alloy wire shall be selected for chemical analysis.
2/ A lot shall be rejected if any of the samples fail.
3/ The average of 5 specimens shall be reported and used for determining compliance with requirements.

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4.3 Tests.

4.3.1 Specimen preparation. Specimens for ultimate tensile strength and elongation tests shall, when required, be given the applicable heat treatment specified in 4.3.1.1 and 4.3.1.2 before testing.

4.3.1.1 Softening heat treatment. The specimens selected from each lot or control number shall be placed in a furnace at 1652° F for 10 minutes and quenched in water at room temperature.

4.3.1.2 Hardening heat treatment. The specimens selected from each lot or control number which have previously been given the heat treatment described in 4.3.1.1, shall be placed in a furnace at 840° F. After being held at 840° F. for 2 minutes, the temperature shall be uniformly reduced to 480° F. in 30 minutes, and the specimens quenched in water at room temperature.

4.3.2 Ultimate tensile strength. Determinations for ultimate tensile strength shall be made on 5 hardened specimens (see 4.3.1.2) selected from each lot or control number. The load shall be applied slowly and continuously at a loading rate of 1 inch per minute until rupture occurs. The average of the 5 specimens shall be reported and used for determining compliance with specification requirements.

4.3.3 Elongation. Determination of elongation shall be made on: (a) 5 specimens as received from the manufacturer, (b) 5 softened specimens (see 4.3.1.1) and (c) 5 hardened specimens (see 4.3.1.2). The gage length shall be 2 inches. The load shall be applied slowly and continuously at a loading rate of 1 inch per minute until rupture occurs. The determined value shall be recorded to the nearest 0.5 percent. Specimens which break outside the gage marks may be repulled. The average of 5 specimens (5 each of conditions (a), (b), and (c) above) shall be reported and used for determining compliance with specification requirements.

4.3.4 Chemical analysis. The samples selected in accordance with 4.2.2 shall be analyzed in conformance with methods 111 or 112 of Federal Standard No. 151. In case of dispute, method 111 shall be used for referee analysis.

4.3.5 Weight. Weighing of individual units of issue and of each delivery to Government shall be performed on a precision balance accurate to 0.001 g for individual weighing and accurate to 0.01g for each delivery to the Government.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.1).

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5.1.1 Unit of issue. Each four pennyweight of wire in not more than two lengths, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each unit of issue shall be packaged straight (not coiled) in a plastic container. Closure shall be adequate to prevent accidental opening.

5.1.2.2 Intermediate package. Five unit packages shall be packed in a box conforming to PPP-B-566 or PPP-B-676. Closure shall be as specified in the appendix of the applicable box specification.

5.1.3 Level C. Each four pennyweight of wire in not more than two lengths shall be packaged in a manner that will afford adequate protection against physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.1).

5.2.1 Level A. Fifty intermediate packages (250 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping, and waterproofing shall be as specified in the appendix of the box specification.

5.2.2 Level B. Fifty intermediate packages (250 units) shall be packed in an exterior container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged wire shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier's rules and regulations.

5.3 Marking.

5.3.1 Interior packages. Each unit and intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number, lot (control) number, and name of contractor shall be shown.

5.3.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129. The lot (control) number shall be shown. The item description, item identification, item name, and trade or brand name shall not be shown on the exterior container.

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6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.3 National Stock Number (NSN) coverage. This specification covers the following item appearing in the Federal Supply Catalog:

NSN	Item Identification
6520-00-580-6950	WIRE, NONELECTRICAL, Platinum Alloy, Round, 4 Pennyweight, 18 Gage

Military Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
DLA-DM

Project No. 6520-1435

Review activity:
DoD - MB

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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